

Making Every Experiment Count

A cheap guide for expensive tests

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The Problem: Few Expensive Tests

Your job: tune parameters, but you face:

- Only about **fifteen expensive** experiments allowed
- **Many interacting** parameters — more than ten
- Change one, the others move — can't tune one at a time
- So: spend a tiny budget in the **smartest way**?

Why It Is Hard: Black Box & Budget

- **Black box:** you see input and output, never the inside
- **Budget:** only fifteen precise runs, but a huge space
- Task: find the **best point** of a high-dimensional black box

The Key Idea: Two Channels

Good news — our black box has two channels:

$$h_k(\mathbf{x}) = h_{k-1}(\mathbf{x}) + \delta_k(\mathbf{x})$$

- **Cheap** channel: run many times, but **biased**
- **Expensive** channel: only 15 runs, accurate
- Idea: combine — accurate = cheap + a correction

The Skeleton Comes from One Person



Figure 1: V. R. Joseph

- V. R. Joseph, Georgia Tech
- **Maxpro** design — where to start
- **Sequential** design — add points one by one
- **Fusion** (KOH) — join cheap and expensive data

Tool 1: Maxpro Design

Maxpro criterion:

$$\min_D \sum_{i=1}^n \prod_{j \neq i} \frac{1}{\sum_{k=1}^p (x_{ik} - x_{jk})^2}$$

- Start **evenly**: spread out in **every** projection

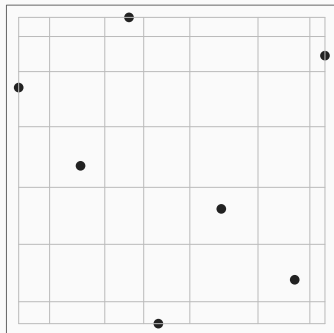


Figure 2: Maxpro design

Tool 2: The Stand-In Model (Gaussian Process)

- A **Gaussian process** is a normal distribution over **whole functions** — not one number
- It predicts a value **and** an **uncertainty** at every point
- Uncertainty tells us **where to test next**: hopeful & unsure

The Whole Pipeline

1. **Spread out** the starting points
2. Feed in many **cheap** runs $\rightarrow$ a **rough map**
3. Send **one expensive** run to the predicted valley
4. **Repeat** — the map gets sharper every round

$$\hat{\mathbf{x}} \approx \arg \min_{\mathbf{x} \in [0,1]^p} h_{\text{H}}(\mathbf{x})$$

- Goal: minimize the expensive function with only **15 precise runs**
- Three small questions:
 - Does **combining the channels** help?
 - Does **screening** the parameters help?
 - Does it still work **in high dimensions**?

Answer 1: Combination Is the Key

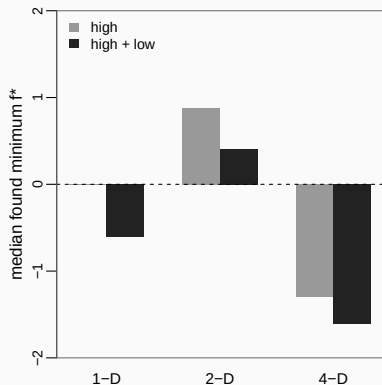


Figure 3: Combination is the key

- Metric: **median** of the best found over 20 repeats
— lower is better
- Cheap + precise **beats** precise alone
- 1-D: precise alone stuck at 0; with fusion → **global optimum**
- 2-D: lands right next to the **known answer**

Answer 2: Screening Is Useless

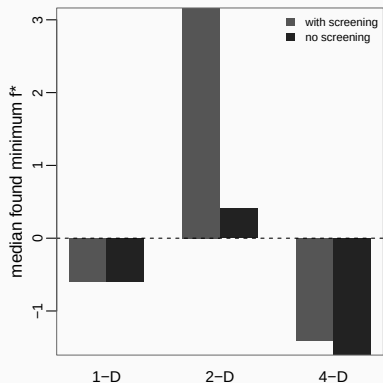


Figure 4: Screening is useless

- Screening **rank**s parameters by cheap runs
- But cheap runs are **biased**
→ ranking is unreliable
- Result: **removing** screening actually helps

Answer 3: High Dimensions Fail

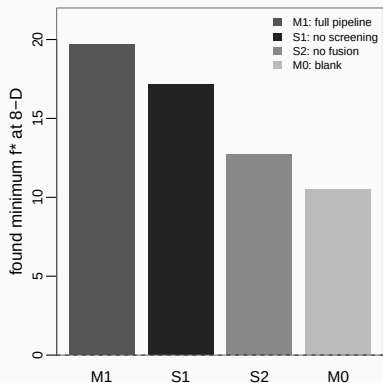


Figure 5: High dimensions fail

- In 8-D, fusion **collapses** too
- Too few precise points to **sketch an 8-D landscape**
- Fusion is decisive — but it has a **ceiling**

The Whole Picture

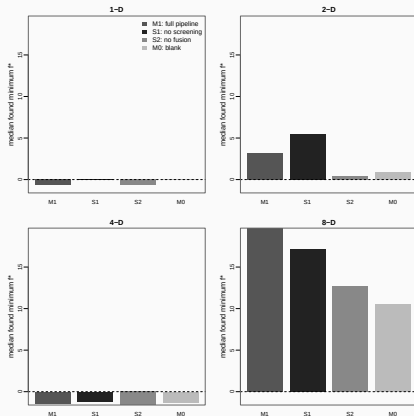


Figure 6: All results

- Low dims: full pipeline wins; drop fusion $\rightarrow$ breaks
- High dims: **nothing works**
- One line: **make every precise test count;**
limit = dimensionality

My Future Directions:

1. **When is screening trustworthy?** — find the bias conditions
2. **Tame high dimensions?** — add structure, or reduce dimension first
3. **Real-world tool:** give it a budget — get the best plan

Summary

- Not “more precise runs” — but “**more value per run**”
- Limit: the **curse of high dimensions**

Thank you!